ROSHAN IMMANUEL E

+918438664587 | roshanimmanuel.work@gmail.com

LinkedIn: www.linkedin.com/in/roshan-immanuel-e-1b1aa4353 Github: https://github.com/roshanimmanuel792

Portfolio: https://os-portfolio-one.vercel.app

EXPERIENCE

IAESTE LC KARUNYA - *Junior Coordinator* Feb. 2026 – Present
Coimbatore, India

- Coordinated and supported students applying for international technical internships, guiding them through documentation, applications, and exchange procedures.
- Managed communication between applicants, local committees and National committees to ensure smooth processing of outgoing exchanges.

E-CELL - *Corporate Relations* Aug. 2025 – Oct 2025
Coimbatore, India

- Facilitated corporate outreach and partnership development by engaging with industry representatives to support E-Cell initiatives, events, and sponsorship opportunities.
- Coordinated with startups, companies, and internal teams to strengthen industry–academia collaboration and ensure successful event execution.

PROJECTS

TrekGuardian – https://github.com/roshanimmanuel792/Trek-Guardian.git

- Developed Trek Guardian**,** an **IoT+ML** based safety system using ESP8266 and multi-sensor integration (MAX30102, BME280/HW611, DHT11, NEO-6M GPS, OLED, SD module) to continuously monitor SpO_2, heart rate, altitude, and environmental conditions in real time.
- Built a **machine learning** pipeline (Decision Tree Classifier in Python**)** trained on physiological and altitude datasets to predict altitude sickness risk, exporting the model rules to embedded firmware for on-device inference.
- Implemented dual-mode emergency communication**:** cloud-based alerts via WiFi and an offline captive portal hotspot that displays user vitals to nearby devices when network connectivity is unavailable.

LikeCircle: **Live Demo** https://like-circle-v1.vercel.app

- Built a full-stack real-time discussion platform using React 19, TypeScript, Express.js, and PostgreSQL, implementing **Socket.IO** real-time messaging with polling fallback, threaded conversations, and **900K+** unique anonymized alias combinations for privacy-first interactions.
- Implemented secure authentication and API architecture using OAuth 2.0 Google Sign-In, JWT access/refresh tokens, and RESTful APIs, with **CORS protection**, token refresh mechanisms, and logout revocation for secure session management.
- Optimized performance and user experience with Tailwind CSS responsive UI, Vite build tooling, database indexing, and connection pooling, enabling **sub-100ms** API responses and real-time activity tracking.

CropVision: https://github.com/roshanimmanuel792/Crop-Vision-v1.git

- Built and Developed full-stack disease detection application using **Java 17, Spring Boot 3.5.7, RESTful API,** Spring Data JPA, Hibernate ORM, and MySQL; engineered responsive JavaFX 21.0.2 GUI with entity mapping, connection pooling, and multipart form data handling via Apache HttpClient 5, Maven, and repository pattern implementation. •
- Implemented layered N-tier architecture with dependency injection, MVC pattern, and **automatic schema generation**; demonstrated proficiency in **ORM database design**, HTTP protocol, JSON processing, end-to-end feature implementation, and client-server communication for production-ready full-stack development.

EDUCATION

Karunya Institute of Technology And Sciences Aug. 2024 – May 2028
Bachelor of Technology in Computer Science and Engineering 2nd Year(8.56 CGPA) Coimbatore, India

SKILLS

Languages : Python, React.js,Typescript.js,Express.js,Tailwind CSS, SQL (PostgreSQL, MySQL),React,ML,Java,SpringBoot
Tools : Unix,Git,Github,Figma,API's,TUI/CLI,Twillio,Firebase.